Probability Midterm 1: Makeup

50 minutes • 100 points • Six problems

Grading note. Award method credit after an earlier error; do not deduct twice for the same mistake. Suggested working times for Problems 1–6 are 5, 5, 6, 8, 10, and 9 minutes, leaving 7 minutes to check.
Instructions. Answer all six problems and show your reasoning. Exact answers are acceptable; binomial coefficients need not be evaluated. No calculator is needed. Use the spaces provided; clearly label any work continued on the back. In Problem 6, use the guided steps rather than quoting a coupon-collector formula. Reserve time to check your work.

1. Sets — 10 points

Let $U = \{1, 2, \ldots, 12\}$, $A = \{1, 3, 5, 7, 9, 11\}$, and $B = \{1, 2, 3, 4, 5\}$. Take all complements relative to U.

(a) Find $A \cap B$, $A \setminus B$, and $(A \cup B)^c$. (6 points)

Solution.

$$A \cap B = \{1, 3, 5\}, \quad A \setminus B = \{7, 9, 11\}, \quad (A \cup B)^c = \{6, 8, 10, 12\}.$$

Credit: 2 per correct set; allow partial credit for a single entry error.

(b) For these sets A, B, is the following statement true for every $C \subseteq U$? Justify with a proof or a counterexample. (4 points)

$$C \subseteq A \cup B \quad \Longrightarrow \quad C \subseteq A \text{ or } C \subseteq B.$$

Solution. False: take $C = \{2, 7\}$. Both elements belong to $A \cup B$, but $2 \notin A$ and $7 \notin B$. Hence neither containment on the right holds.

Credit: False: 1; valid example: 1; union containment: 1; both failed containments: 1. Accept any valid counterexample.

2. A probability space — 10 points

Let $\Omega = \{a, b, c, d\}$, with every subset an event. The probabilities of a, b, c, d are $2k, 3k, 4k, 5k$, respectively.

(a) Find k, explaining the probability axiom you use. (4 points)

Solution. The disjoint singletons partition Ω. Additivity and $\mathbb{P}(\Omega) = 1$ give $14k = 1$, so $k = 1/14$. The probabilities are nonnegative.

Credit: Normalization/additivity explanation: 2; equation: 1; value: 1.

(b) Let $E = \{a, c\}$ and $F = \{b, c\}$. Find $\mathbb{P}(E)$, $\mathbb{P}(E \cap F)$, and $\mathbb{P}(E^c)$. (6 points)

Solution. Since $E \cap F = \{c\}$ and $E^c = \{b, d\}$,

$$\mathbb{P}(E) = \frac{6}{14} = \frac{3}{7}, \qquad \mathbb{P}(E \cap F) = \frac{4}{14} = \frac{2}{7}, \qquad \mathbb{P}(E^c) = \frac{8}{14} = \frac{4}{7}.$$

Credit: 2 per probability. Carry forward an earlier error in k: accept the correctly obtained expressions $6k, 4k, 8k$.

3. Inclusion–exclusion **15 points**

In a group of 45 students, 20 play tennis, 17 play basketball, and 13 play volleyball. Eight play both tennis and basketball, six both tennis and volleyball, and five both basketball and volleyball. Three play all three. **The pairwise counts include students who play all three sports.**

(a) How many students play at least one of these sports? Use inclusion–exclusion. (11 points)
Solution. Writing T, B, V for the three sets,

$$\begin{aligned}|T \cup B \cup V| &= |T| + |B| + |V| - |T \cap B| - |T \cap V| - |B \cap V| + |T \cap B \cap V| \\ &= 20 + 17 + 13 - 8 - 6 - 5 + 3 = 34.\end{aligned}$$

Credit: Individual counts: 3; subtracting pairwise intersections: 3; restoring the triple intersection: 3; result: 2.

(b) How many students play none of these sports? (4 points)
Solution. There are $45 - 34 = 11$ such students.
Credit: Complement subtraction: 3; result: 1. Carry forward an earlier error.

4. Balls in an urn **20 points**

An urn contains six red, three blue, and four green balls. Four balls are selected uniformly without replacement: every four-ball subset of the thirteen individual balls is equally likely.

(a) Find the probability of selecting exactly two red, one blue, and one green ball. (10 points)
Solution. Choose the required number of each color and divide by the total number of four-ball subsets:

$$\mathbb{P}(\text{two red, one blue, one green}) = \frac{\binom{6}{2}\binom{3}{1}\binom{4}{1}}{\binom{13}{4}} = \frac{180}{715} = \frac{36}{143}.$$

Credit: Uniform denominator: 3; red choice: 2; blue and green choices: 2; product and ratio: 3. No numerical simplification needed.

(b) Find the probability of selecting at least one green ball. (10 points)
Solution. The nine non-green balls give the complementary count:

$$\mathbb{P}(\text{at least one green}) = 1 - \frac{\binom{9}{4}}{\binom{13}{4}} = 1 - \frac{126}{715} = \frac{589}{715}.$$

Credit: Complement event: 3; favorable complement count: 3; denominator: 2; subtracting from 1: 2. Accept an equivalent direct count.

5. A density, its mean, and its variance **25 points**

A continuous random variable X has probability density

$$f(x) = \begin{cases} cx^2, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant. Show the integrals you use.

(a) Find c. (5 points)

Solution. Normalization requires

$$1 = c \int_0^1 x^2 \, dx = c \left[\frac{x^3}{3} \right]_0^1 = \frac{c}{3}.$$

Thus $c = 3$, which also makes the density nonnegative.

Credit: Normalization integral: 2; integral value: 2; constant: 1.

(b) Compute $\mathbb{E}[X]$. (8 points)

Solution.

$$\mathbb{E}[X] = \int_0^1 x(3x^2) \, dx = 3 \left[\frac{x^4}{4} \right]_0^1 = \frac{3}{4}.$$

Credit: Moment integral including the factor x: 4; antiderivative and limits: 3; result: 1. Carry forward an earlier value of c.

(c) Compute $\mathbb{E}[X^2]$, and hence $\mathrm{Var}(X)$. (12 points)

Solution.

$$\mathbb{E}[X^2] = \int_0^1 x^2(3x^2) \, dx = 3 \left[\frac{x^5}{5} \right]_0^1 = \frac{3}{5}.$$

Therefore

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{3}{5} - \left(\frac{3}{4} \right)^2 = \frac{48 - 45}{80} = \frac{3}{80}.$$

Credit: Second-moment integral: 4; evaluation: 3; variance identity: 3; substitution and arithmetic: 2. Accept a correct direct variance integral.

6. Collecting five sticker types **20 points**

Each packet contains one sticker. There are five sticker types, each equally likely, and different packets have independent contents. Duplicates are possible. Starting with no stickers, let T be the number of packets opened up to and including the first packet that completes the collection of all five types.

(a) After exactly k distinct types have been collected, let p_k be the probability that the next packet contains a new type. Find p_0, p_1, p_2, p_3, p_4, and explain your reasoning. (6 points)

Solution. There are $5 - k$ missing types out of five equally likely types, so

$$p_k = \frac{5-k}{5}, \qquad (p_0, p_1, p_2, p_3, p_4) = \left(1, \frac{4}{5}, \frac{3}{5}, \frac{2}{5}, \frac{1}{5}\right).$$

Credit: Missing-type counting argument: 1; five correct probabilities: 1 each.

(b) Let W_k be the number of additional packets needed to go from exactly k collected types to $k+1$ collected types, including the packet containing the new type. Find $\mathbb{E}[W_k]$ for $k = 0, 1, 2, 3, 4$. (8 points)

You may use: A geometric waiting time that counts trials through the first success, with success probability p, has mean $1/p$.

Solution. Until a new type appears, the missing set stays fixed. Independent packets therefore give a geometric wait with success probability p_k (the first wait is deterministically one). Thus

$$\big(\mathbb{E}[W_0], \mathbb{E}[W_1], \mathbb{E}[W_2], \mathbb{E}[W_3], \mathbb{E}[W_4]\big) = \left(1, \frac{5}{4}, \frac{5}{3}, \frac{5}{2}, 5\right).$$

Credit: Applying the geometric reciprocal rule: 3; five means: 1 each. Carry forward earlier probabilities when reciprocals are taken consistently.

(c) Write T as a sum of the waiting times and find $\mathbb{E}[T]$. Name the expectation rule used. (6 points)

Solution. The stages partition all packets opened, so $T = \sum_{k=0}^{4} W_k$. Linearity of expectation gives

$$\mathbb{E}[T] = 1 + \frac{5}{4} + \frac{5}{3} + \frac{5}{2} + 5 = \frac{137}{12}.$$

Independence of the waiting times is not required for linearity.

Credit: Stage decomposition: 2; linearity and sum: 2; evaluation: 2. Accept the exact unevaluated sum.

End of paper. Check that you have answered all six problems.